MDT Ã— GNN â€” decision points (2026-07-24)

Two pre-registered rounds, 560+ grid rows, 8 intact graph datasets, verdicts emitted by code. Full record: `GRAPH_MDT_RESEARCH_LEDGER.pdf`, `results/graph_mdt/`.

Headline results

1. **Given identical inputs, no GNN beats MDT consensus.** GCN/SAGE Ã— GAE/BGRL, effects -0.07 to -0.30 AMI vs MDT-with-graph-views, wins â‰¤2/6 primary datasets. GNN wins appear only where the pre-run screen showed weak graphs â€” exactly where MDT's diffusion assumption also hurts.
2. **Zero-training MDT reaches published GNN territory** once given the same graphs: DBLP-7907 NMI 0.59 (HDMI 0.582, BTGF 0.624), ACM 0.63 (O2MAC 0.69). Caveat for a paper: protocols differ across lineages; numbers need aligned reruns, not table copying.
3. **Per-node view gating has leverage but the heuristic signal is wrong.** Feature-anchored gates lift ACM to ACC 0.886 / NMI 0.685 (exploratory, O2MAC-level) yet collapse DBLP. Validation on untouched amazon was uninformative (no method above 0.05 AMI there), so the ACM number stays exploratory; the gate's direction behaved exactly as the feature-anchoring model predicts across all 9 datasets.
4. **The GNN's only surviving unique gain:** +0.05 AMI on minesweeper (structure-only synthetic control); a spectral AÂ² view does not close it.

Decisions needed

1. **Publication route.** Recommend: new experiments section in the mdt_ddm paper (multiplex benchmarks strengthen the lab method) rather than a standalone negative-results paper. Needs protocol-aligned baseline reruns.
2. **GNN thread.** Closed under the frozen rules. Only licensed reopening: a new pre-registration with cross-view neighbourhood agreement as the gate signal (~1 week). Worth it or not?
3. **Lab tooling fixes independent of any paper:**
4. silhouette-based trajectory selection prefers collapsed embeddings â€” replace with modularity-on-graph criterion;
5. Gaussian-euclidean kNN collapses on BoW features (homophily 0.36 vs 0.71 cosine) â€” affects existing MDT pipelines;
6. random-trajectory consensus beats single-trajectory selection.

Methodology guarantees (why these numbers can be trusted)

Pre-registered gates and thresholds; label-free model selection everywhere; every amendment timestamped and evidence-linked before affected rows existed; exploratory results explicitly denied claim status; verdict labels computed only by `verdicts.py`, never written by hand.